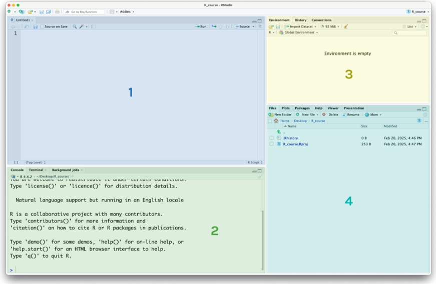
Session 1: Introduction to R

Summary sheet

RStudio



& don't forget to create an **R project** for each new analysis!

1. Script: write, run & save code
   to run code: command/control + enter
2. Console: see results, run quick code
3. Environment: variables
4. Files/plots/help

Comments

Comments start with a **hashtag** & are ignored by R. Use them to add notes to your code `# just like this! :)`

Maths

addition	+	division	/
subtraction	-	exponents	^
multiplication	*		

Brackets () can be used to specify order of operations, e.g. `(1 + 3) * 2`

Comparisons

equal to	`==`
not equal to	`!=`
less/greater than	`>` or `<`
less/greater or equal to	`<=` or `>=`

Give logical (`TRUE`/`FALSE`) as output

Variables

Use `<-` to assign, e.g. `x <- 10`

Naming variables:

- **no** spaces
- **no** special characters (e.g. `! $ &`)
- **no** starting with number
- **avoid** capital letters
- **avoid** names that aren't descriptive

Functions

function name ↓   **comma** separates arguments ↓

```
round(3.1459, digits = 2)
```

surrounded by brackets

Get help with `?function_name`

Atomic data types

logical (true/false)	`TRUE`
numeric (numbers)	`1000`
character (text)	`"hello"`

Check type with the `class()` function

Data structures

Vector   Matrix   Data frame / tibble

Must be same type   *Different types OK*

Create vectors with `c()` e.g. `c(1,2,3)`

Create data frames from a series of vectors with `data.frame()`

e.g.
```
data.frame(
    row1 = c(1, 2),
    row2 = c("a", "b"))
```

See just the start with `head()`, just the end with `tail()` or the open the whole thing with `View()`

R is a vectorised language, so many functions/operators work on vectors directly e.g. `c(1, 2, 3) + 10`

Some functions are designed for vectors e.g. `mean()` or `range()`

Subsetting

Access columns from a data frame using `$` e.g. `df$col_1`

This can also be used to create new columns with assignment operator:
`df$new_col <- c(1, 2, 3)`

For vectors (or data frame columns), use `[]` to access specific elements, e.g. `letters[5]` or use a colon `:` for a range, e.g. `letters[1:5]`

Paths

folder names   file name

`"my_analysis/data/genes.csv"`

/ to separate   file extension